

Mediterranean Sea is connected to the Atlantic Ocean and is almost completely enclosed by land. It is surrounded by African continent in south, Europe in the north and Asia in the east. This sea is connected to the Atlantic Ocean by Gibraltar Strait. The Bosphorus Strait connects the Black Sea to the sea of Marmara and thence to the Aegean and Mediterranean Sea via the Dardanelles Strait. Red sea by Suez canal. The countries bordering Mediterranean Sea are - Albania, Algeria, Cyprus, Greece, France, Egypt, Montenegro, Slovenia, Spain, Syria, Lebanon, Israel, Italy, Libya, Malta, Morocco, Tunisia, Turkiye, Croatia, Bosnia and Herzegovina and Monaco. Therefore, option (c) is the correct answer.

21. Which one of the following countries is not situated along the coast of Baltic Sea?

- (a) Latvia and Poland (b) Latvia and Russia
(c) Sweden and Russia (d) Sweden and Norway

U.P. R.O./A.R.O. (Mains) 2021

Ans. (d)

The Baltic sea is surrounded by 9 countries in Northern Europe. The countries located on its coast are - Denmark, Germany, Poland, Lithuania, Latvia, Estonia, Russia, Finland and Sweden. So it is clear that Norway is not located on the coast of the Baltic Sea. Thus the correct option is (d).

22. Consider the following statements:

- 1. The Red Sea receives very little precipitation in any form.**
- 2. No water enters the Red Sea from rivers.**

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2024

Ans. (c)

The Red Sea is a sea inlet of the Indian Ocean, lying between Africa and Asia. Its connection to the ocean is in the south, through the Bab-el-Mandeb Strait and the Gulf of Aden. The Red Sea region receives very little precipitation in any form, although prehistoric artifacts indicate that there were periods with greater amounts of rainfall. No water enters the Red Sea from rivers, and rainfall is scant; but the evaporation loss—in excess of 80 inches per year—is made up by an inflow through the eastern channel of the Bab-el-Mandeb Strait from the Gulf of Aden. The Red Sea is one of the saltiest bodies of water in the world, owing to high evaporation and low precipitation; no significant rivers or streams drain into the sea, and its southern connection to the Gulf of Aden, an arm of the Indian Ocean, is narrow. Hence, both statements 1 and 2 are correct.

The Ocean Currents

*An ocean current is a continuous directed movement of sea -water. Surface ocean currents are driven by the prevailing winds and by the rotation of the Earth. Deep sea currents, upper 100 meters below the ocean surface are driven by differences in water temperature and salinity, which have an impact on the density of deep water and on its movement. The main causes of ocean currents are.

- (i) Differential heating of the equatorial and polar oceans.
- (ii) Trade and anti-trade winds
- (iii) Coriolis effect
- (iv) Variation in salinity and temperature of the oceans.
- (v) Configuration of the coastline and ocean floor.

*The oceanic currents play a huge role in regulating the planet's temperature.

*Based on the temperature gradient, the ocean currents can be categorized into two types - **warm ocean currents** and **cold ocean currents**.

Factors Responsible for Change in Direction of Ocean Currents-

- (i) **Configuration of the coast** - When a landmass lies perpendicular to the prevailing current, the obstruction thus caused deviates the current to move parallel to the landmass.
- (ii) **Ocean floor reliefs** - The diversified ocean floor with distinctive reliefs can also affect the oceanic currents and their direction.
- (iii) **Seasonal Variations** - for example, the monsoonal winds change the direction of the ocean currents.
- (iv) **The Earth's Rotation** - The Earth's rotation deflects freely moving objects including ocean currents, to the right. In the northern hemisphere, this is a clockwise direction (e.g. the circulation of the Gulf Stream Drift and the Canary Current). In the southern hemisphere it is an anti-clockwise direction (e.g. the Brazilian Current and the West Wind Drift).

1. Currents of Atlantic Ocean - *Under the influence of trade winds the North Equatorial Warm Current flows east to west from African coast in the east to the West Indies.

***Antilles Current** - Near east coast of South America the North Equatorial Current splits into two branches. One branch flows northward, east of the Antilles (West Indies) as Antilles Current and it joins the Florida Current past the

outer Bahamas. The other branch turns north and flows with the North Equatorial Current into the Caribbean Sea and the Gulf of Mexico.

***Florida Warm Current** - Under the influence of the trade winds the current flows from the Gulf of Mexico towards the Strait of Florida. Here it adjoins the Antilles Current.

***Gulf Stream** - When the Florida warm current and Antilles warm current join together, they are collectively known as the Gulf Stream. The Gulf Stream is a warm and swift Atlantic Ocean current that follows the eastern coastline of the United States.

***Canary Cold Current** - It flows southwestward along the northwest coast of Africa as far south as Senegal before turning westward to join the North Equatorial Current.

***East Greenland Current** - The current travels along the eastern coast of Greenland in North-South direction.

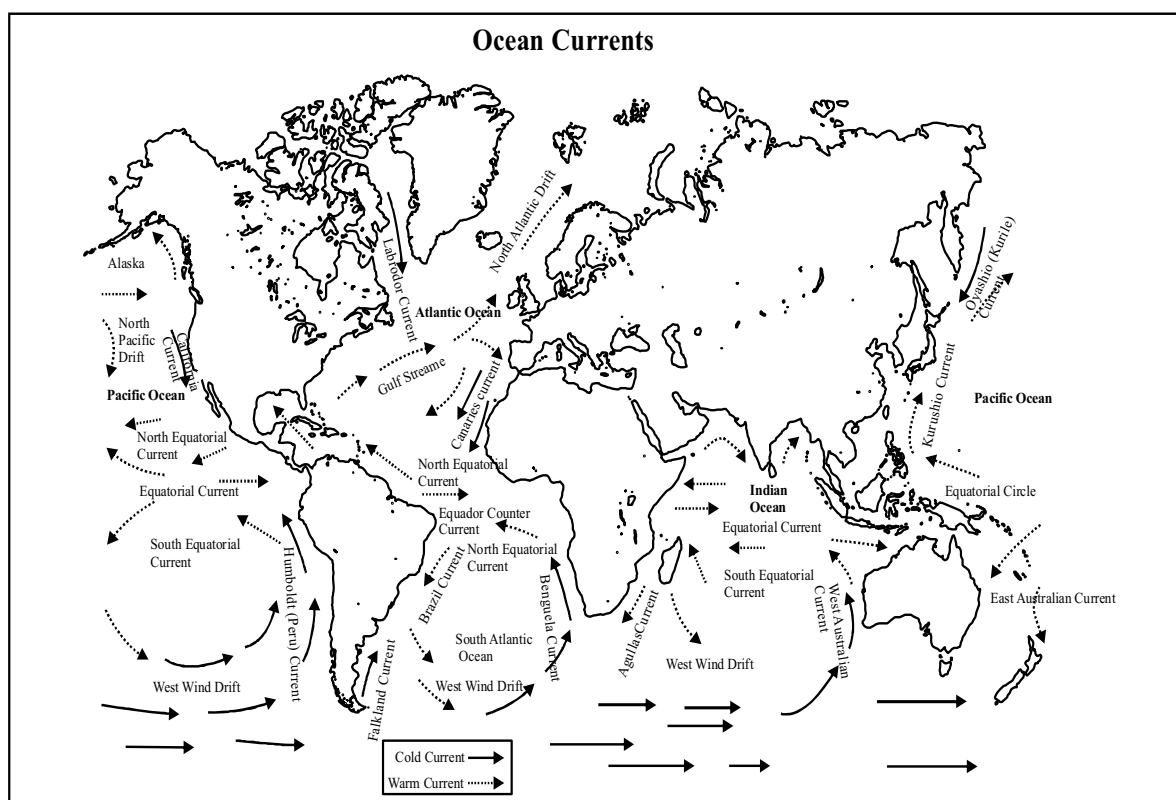
***Labrador Cold Current** - The current flows from Baffin Bay and the Davis Strait and travels along the coast of Labrador in a north-south direction.

***North Atlantic Drift/Current** - The current originates between 45° North Latitude and 45° West Longitude. This current is responsible for carrying the Gulf Stream to Europe. Countries like Norway which are located in a sub-polar region

benefit the most from this drift of current. This is because during winters northern Europe receives heavy snowfall, thus the warm North Atlantic Drift/Current prevents extreme cold in the region and also keeps the harbors ice-free. As a result, the North Atlantic Warm Drift/Current is also known as the 'Blanket of Europe'.

Brazilian Current is a warm current. At 40° South Latitude, the **Falkland Current** meets the Brazilian Current. Collectively they flow eastward as South Atlantic Drift. upon reaching the west coast of Africa the current is diverted northwards as the cold **Benguela Current**. It brings the cold polar waters of the west drift into tropical latitudes. Driven by the regular southeast trade winds, the Benguela Current moves equatorward in a northwesterly direction to join the South Equatorial Current. This completes the circulation of the currents in the South Atlantic.

2.Currents of Indian Ocean - As in the other oceans, the currents of the South Indian Ocean form a circuit. The South Equatorial Current turns Southward and divides into the Madagascar Current and Mozambique Current. These currents merge as Agulhaing Warm Current with the West Wind Drift, flowing eastwards and turning equatorward as the West Australian Current (cool current). In the North



Indian Ocean, there is a complete reversal of the direction of currents between summer and winter, due to the changes in monsoon winds. In summer from June to October, when the dominant wind is the southwest Monsoon, the currents are blown from a south - westerly direction as the **southwest Monsoon Drift**. This is reversed in winter, beginning in December, when the northeast Monsoon blows the currents from the north-east as the north-east Monsoon Drift. The currents of the Monsoon Drift.

3. Currents of Pacific Ocean - The pattern of circulation in the Pacific is similar to that of the Atlantic except for modifications that can be expected from the greater size and more open nature of the Pacific.

*The North Equatorial Current is a current in the Pacific Ocean, originating in the Mid Americas and flowing east to west. The Northeast Trade Winds blow the North Equatorial Current to the coast of the Philippines.

***Kuroshio Warm Current** - After reaching the island of Philippines the North Equatorial Current flows towards Taiwan and the northwards past Japan. Here, the current is known as Kuroshio Current.

***Californian Cold Current** - When the North Pacific Drift reaches the western coast of North America it gets bifurcated into two parts. One travels northwards and the other travels southwards along the western coast of North America. The southward-moving current is known as the Californian current.

***Alaskan Stream Current** - The northward branch of the North Pacific current when it enters the Gulf of Alaska is known as Alaskan warm current. Again it flows along the coast of Alaska, where it is known as the Alaskan stream, which is a cold stream

***Oyashio Cold Current** - The current originates in the Bering Strait and travels in a northsouth direction near the eastern coast of Kamchatka Peninsula.

***Kurile Current** - Passing through the Bering Strait and transporting cold water from the Arctic Sea into the Pacific Ocean and the **Sea of Okhotsk**. Again passing through the east coast of Sakhalin Island it is joined by the Okhotsk current to meet the warm Japan Current as of the **Oyashio**.

***East Australian Warm Current** - The East Australian current sweeps warm water down the east coast of Australia.

***Peruvian Cold Current** - Also known as the Humboldt Current, travels along the west coast of South America.

***Equatorial Counter Current** - Due to North and South Equatorial Currents the Pacific Ocean accumulates a huge amount of water on both sides of the equator. This results in an imbalance in Ocean's water levels. To current this imbalance equatorial counter currents come into action.

1. Consider the following factors

1. **Rotation of the Earth**
2. **Air pressure and wind**
3. **Density of ocean water**
4. **Revolution of the Earth**

Which of the above factors influences the ocean currents?

- | | |
|-------------|----------------|
| (a) 1 and 2 | (b) 1, 2 and 3 |
| (c) 1 and 4 | (d) 2, 3 and 4 |

I.A.S. (Pre) 2012

Ans. (b)

Ocean current is a continuous, directed movement of sea-water generated by forces such as rotation of the Earth, wind, temperature and salinity difference, air pressure, sea water density differences etc. Revolution of the Earth does not influence the ocean currents. Hence, option (b) is the correct answer.

2. Consider the following statements:

1. **Ocean currents are the slow-surface movement of water in the ocean.**
2. **Ocean currents assist in maintaining the Earth's heat balance**
3. **Ocean currents are set in motion primarily by prevailing winds**
4. **Ocean currents are affected by the configuration of the ocean.**

Which of these statements are correct?

- | | |
|----------------|-------------------|
| (a) 1 and 2 | (b) 2, 3 and 4 |
| (c) 1, 3 and 4 | (d) 1, 2, 3 and 4 |

I.A.S. (Pre) 2002

Ans. (d)

The flow of ocean water in a particular direction is called the oceanic current. Mostly ocean currents are set in motion by the prevailing wind. If the wind is blowing, normally the surface water is in motion. Thus, statements 1 and 3 are correct. In cold regions hot streams (Gulf Stream) and in hot regions cold streams play their role in heat stabilization. Thus statement 2 is also correct. Ocean currents are also affected by the configuration of the ocean besides prevailing winds and Earth's rotation. Thus, statement 4 is also correct.

3. Which one of the following factors is responsible for the change in the regular direction of the ocean currents in the Indian Ocean?

- (a) Indian Ocean is half an Ocean
- (b) Indian Ocean has Monsoon drift

- (c) Indian Ocean is a land-locked ocean
(d) Indian Ocean has greater variation in salinity

I.A.S. (Pre) 1997

Ans. (b)

The ocean currents of the Indian Ocean are slightly different from these of the Pacific and the Atlantic Oceans. Land and monsoon winds have a major impact on the Indian Oceanic current. The direction of current flow changes twice in the Northern Indian Ocean due to the North-East and South-West monsoon winds whereas the Southern Indian Ocean currents have a definite pattern. Hence, option (b) is the correct answer.

4. Agulhas current flows in –

- (a) The Indian Ocean (b) The Pacific Ocean
(c) North Atlantic Ocean (d) South Atlantic Ocean

R.A.S./R.T.S. (Pre) 1999

Ans. (a)

Agulhas current is the warm water current that flows in South Indian Ocean. The South Equatorial current turns southward and is divided into the Madagascar current and Mozambique current. These currents merge south of Madagascar as Agulhas warm current and then it flows up to southernmost end of Africa later mixes with the West Wind Drift by turning East.

5. Which of the following ocean currents is associated with Indian Ocean?

- (a) Florida Current (b) Canary Current
(c) Agulhas Current (d) Kurile Current

U.P. P.C.S. (Pre) 2020

Ans. (c)

See the explanation of the above question.

6. Which of the following is not a cold ocean current?

- (a) Canaries (b) Humboldt
(c) Oyashio (d) Agulhas

U.P.P.C.S. (Pre) 2017

Ans. (d)

See the explanation of the above question.

7. The country benefiting the most out of the North Atlantic Drift is :

- (a) Poland (b) Portugal
(c) Norway (d) Nigeria

R.A.S./R.T.S.(Pre) 1999

Ans. (c)

Northern Atlantic currents generate near 45° north latitude and 45° west meridian. They carry the warm water of the Gulf Stream towards, Europe, where Norway, situated in the sub-polar region, benefits most from this current. This benefit is such that at the onset of winter chances of most countries in Europe coming under the expansion of ice-cover become greater as the warm North Atlantic currents block the expansion of ice-cover. Therefore this current is also known as 'Blanket of Europe'.

8. Which one of the following is a cold current of the South Atlantic Ocean?

- (a) Canary current
(b) Benguela current
(c) Agulhas current
(d) Brazil current

U.P.P.C.S. (Pre) 2005

Ans. (b)

An only cold current of South Atlantic Ocean is the "Benguela current". It flows along the west coast of Southern Africa towards the north direction. Moving ahead, it merges with South Equatorial current. Therefore option (b) is correct.

Other currents of the options are as follows –

Brazil Current : It is a warm water current that flows along the East coast of Brazil from north to south.

Canary Current : It is a cold water current that flows in the North Atlantic Ocean along the west coast of north-west Africa from north to south.

Agulhas Current: It is a warm water current that flows in South-Indian Ocean from north to south.

9. Which of the following ocean currents does not belong to the Indian Ocean?

- (a) Agulhas Current
(b) Mozambique Current
(c) South Indian Ocean Current
(d) Benguela Current

U.P.P.C.S. (Mains) 2014

Ans. (d)

Agulhas Current flows along the west border of the South Indian Ocean and it is a warm current. Mozambique Current is also a warm current that flows through Mozambique Channel in the West Indian Ocean. In the 40° to 45° latitudes flows the South Indian Ocean current, while Benguela current is a cold water current which flows in South Atlantic ocean along the west coast of Africa.

10. Which one of the following currents does not contribute to forming a complete circle of currents in the South Atlantic Ocean:

- (a) Benguela (b) Brazil
(c) Canary (d) West wind drift

U.P.P.C.S. (Pre) 2000

Ans. (c)

In the South Atlantic Ocean, there exists a fixed order of complete circular currents. This order emerges from South Equatorial Current. After hitting the Brazilian coast, it moves south towards where it is called Brazil current. The disturbance develops due to the rotation of the Earth. At 40° latitude, this current is affected by west wind. It turns east and current coming from the south i.e, Falklands Current meets the Brazil Current. Warm Brazil Current and cold Falklands Current merge and turn eastward as the South Atlantic Current and one of its branches after hitting the west coast of South Africa turns towards the north where it is known as Benguela Current. This current moving ahead and meeting with South Equatorial current forms a complete circle of currents. This complete circle of currents flows in anticlockwise direction. Therefore it is clear that Canary current is not associated with the complete circle of South Atlantic currents. This is actually a cold current that flows in the North Atlantic Ocean.

11. Which of the following oceanic current does not flow in North Atlantic ocean?

- (a) Gulf Stream (b) Benguela
(c) Norwegian (d) Florida

Uttarakhand P.C.S. (Pre) 2021

Ans. (b)

The Benguela is a cold current in the South Atlantic Ocean. The Gulf Stream, Norwegian and Florida are warm currents that flow into the North Atlantic Ocean.

12. The ocean current different from the remaining three is :

- (a) Benguela (b) Brazilian
(c) South Equatorial (d) Peruvian

R.A.S./R.T.S.(Pre) 1999

Ans. (d)

Except for the Peruvian current, all three currents mentioned above are related to the Atlantic Ocean which forms a circle. Peruvian current is related to the Southeast Pacific Ocean. Peruvian current is also known as Humboldt current. It is a cold current that flows along the west coast of the South American continent.

13. Which of the following statements is true?

- (a) Ocean salinity is maximum at the Tropic of Cancer and Capricorn.
(b) Tide comes on the earth every day after exactly 12 hours 30 minutes.
(c) Benguela current is a cold current of the Pacific ocean.
(d) If the Sun, the Earth and the Moon are in a straight line then this situation is the situation of small tide.

U.P.P.C.S. (Pre) 2023

Ans. (a)

Ocean salinity is the term used to define the total content of dissolved salts in seawater. It is usually expressed as parts per thousand (‰) or ppt. Ocean salinity is maximum at the Tropic of Cancer and the Tropic of Capricorn due to low rainfall and high evaporation. Hence statement (a) is correct. The periodical rise and fall of the sea level, once or twice a day, mainly due to the attraction of the Sun and the Moon, is called a tide. Hence statement (b) is incorrect. Benguela current is a cold current but not of the Pacific Ocean. Hence statement (c) is incorrect. When the Sun, the Moon and the Earth are in a straight line, the height of the tide will be higher. These are called spring tides and they occur twice a month, once during the full moon period and another during the new moon period. Hence statement (d) is also incorrect.

14. Which of the following is cold ocean current :

- (a) Peruvian Current (Humboldt Current)
(b) Kuroshio Current
(c) Gulf Stream
(d) Brazil Current

Jharkhand P.C.S. (Pre) 2003

U.P.P.C.S. (Pre) 1995

Ans. (a)

See the explanation of the above question.

15. Which of the following is not matched correctly?

- (a) Brazil Current - South Atlantic Ocean
(b) Humboldt Current - North Pacific Ocean
(c) Gulf Stream - North Atlantic Ocean
(d) Agulhas Current - Indian Ocean

U.P. P.C.S. (Pre) 1998

Ans. (b)

See the explanation of the above question.

16. Gulf Stream is –

- (a) a river in the Gulf
- (b) an oceanic current
- (c) another name of Jet Stream
- (d) a surface wind

44th B.P.S.C. (Pre) 2000

42nd B.P.S.C. (Pre) 1997

Ans. (b)

Gulf Stream is a powerful warm current in the North Atlantic Ocean. It originates at the Gulf of Mexico as the Florida current which merges with the Antilles Current and flows through Cape Hatteras then flows along the east coast of United States of America. Spanish explorer Juan Ponce de Leon first discovered the Gulf Stream in 1513.

17. Which of the following pair of oceanic currents is shown in the given map :



- (a) Benguela and Falkland
- (b) Canary and Humboldt
- (c) Agulhas and Guinea
- (d) Benguela and Guinea

I.A.S. (Pre) 1999

Ans. (d)

In the above map, the current flowing from south to north is Benguela current. It is cold water current. The current flowing from north to south is Guinea Current.

18. Match List-I with List-II and select the correct answer using the code given below the Lists :

List-I	List-II
A. Gulf Stream	1. Pacific Ocean
B. West Wind Drift	2. A slow eastward movement of water over the zone of westerly wind
C. Peru Current	3. Indian Ocean
D. West Australian Current	4. Warm current

Code :

- | | | | |
|-------|---|---|---|
| A | B | C | D |
| (a) 4 | 2 | 1 | 3 |
| (b) 1 | 3 | 4 | 2 |

(c) 4 3 1 2

(d) 1 2 4 3

Jharkhand P.C.S. (Pre) 2013

Ans. (a)

The correctly matched of List-I and List-II is as follows :

Gulf Stream	-	Warm current
West Wind Drift	-	A slow eastward movement of water over the zone of westerly wind
Peru Current	-	Pacific Ocean
West Australian Current	-	Indian Ocean

19. What explains the eastward flow of the equatorial counter-current?

- (a) The Earth's rotation on its axis
- (b) Convergence of the two equatorial currents
- (c) Difference in salinity of water
- (d) Occurrence of the Belt of Calm near the equator

I.A.S. (Pre) 2015

Ans. (b)

There are two equatorial currents flowing from east to west – the north equatorial current and the south equatorial current. In between these two, there flows an equatorial counter current in the opposite direction from west to east. Piling up of water due to the convergence of two equatorial currents gives rise to equatorial counter current. Therefore, it is the Earth's rotation on its axis that is responsible for the eastward flow of equatorial counter current.

20. Which one of the following Oceanic currents is not associated with the Pacific Ocean?

- (a) Canaries
- (b) Kuroshio
- (c) California
- (d) Humboldt

Uttarakhand P.C.S. (Pre) 2016

Ans. (a)

See the explanation of the above question.

21. Which of the following is a warm current?

- (a) Falkland Current
- (b) Brazil Current
- (c) Labrador Current
- (d) Canary Current

U.P. R.O./A.R.O. (Mains) 2017

Ans. (b)

Ocean currents are of two types. Warm Ocean currents— Brazil Current, Gulf Stream, Kuroshio Current, etc. Cold currents - Falkland Current, Labrador Current, Canary Current.